Rising Waters: Flood Prediction System Using Machine Learning

➢ Abstract:

Rising Waters is an AI-based flood prediction system that uses Machine Learning to predict flood risk based on environmental data such as rainfall, humidity, and river water level. The system is developed using Python, Flask, HTML, CSS, and Scikit-learn. It helps users quickly determine whether there is a flood risk by entering input values through a web application.

➢ Problem Statement:

Floods cause severe damage to lives, agriculture, and infrastructure every year. Traditional flood prediction methods may not provide timely or accurate predictions. Therefore, an AI-based system is needed to predict flood risk quickly and assist in disaster preparedness.

➢ Features:

Example

- **Predict flood risk**
- **User-friendly web interface**
- **Machine Learning prediction**
- **Fast processing**
- **Easy to use**
- **Responsive design**

➢ Tech Stack:

What is it?

The technologies, programming languages, frameworks, and tools used to build your project.

Example

Programming Language

- **Python**

Frontend

- **HTML**
- **CSS**

Backend

- **Flask**

Machine Learning

- **Scikit-learn**

Libraries

- **Pandas**
- **NumPy**
- **Job lib**

IDE

- **Visual Studio Code**

➢ Installation Steps:

What is it?

These are the steps another person should follow to run your project.

Example:

1. **Install Python.**
2. **Download the project.**
3. **Open the project in VS Code.**
4. **Install required libraries.**

pip install -r requirements.txt

5. **Run the project.**

python app.py

6. **Open the browser.**

http://127.0.0.1:5000

Usage Instructions:

What is it?

Explains how a user operates your application.

Example

1. **Open the website.**
2. **Enter rainfall.**
3. **Enter humidity.**
4. **Enter river level.**
5. **Click Predict.**
6. **View the prediction result.**

Future Scope:

What is it?

This section explains what improvements can be made in the future.

Example

- **Real-time weather API integration**
- **Mobile application**
- **SMS alert system**
- **IoT sensor integration**
- **Google Maps support**
- **Deep Learning models.**

❖ Methodology

What is it?

The step-by-step process used to develop the project.

Why is it needed?

It explains how the project was built.

Example:

1. **Collect dataset.**
2. **Clean the data.**
3. **Train the ML model.**
4. **Save the model.**
5. **Build Flask application.**

❖ Results and Screenshots

What is it?

Images showing the project in action.

Why is it needed?

It proves that your project works.

Include:

- **Home page**
- **Input page**
- **Prediction result**
- **VS Code project**
- **Folder structure**

✓ Conclusion:

The Rising Waters project successfully predicts flood risk using Machine Learning and provides users with a simple web application for quick predictions.